

Questions on Data Types, Variables and Arrays

Integer and Floating Data Types

1. What is the range of short data type in Java?

- ☐ a) -128 to 127
- ☒ b) -32768 to 32767
- ☐ c) -2147483648 to 2147483647
- ☐ d) None of the mentioned

2. What is the range of byte data type in Java?

- ☒ a) -128 to 127
- ☐ b) -32768 to 32767
- ☐ c) -2147483648 to 2147483647
- ☐ d) None of the mentioned

3. Which of the following are legal lines of Java code?

- 1. `int w = (int)888.8;`
- 2. `byte x = (byte)100L;`
- 3. `long y = (byte)100;`
- 4. `byte z = (byte)100L;`
- ☐ a) 1 and 2
- ☐ b) 2 and 3
- ☐ c) 3 and 4
- ☒ d) All statements are correct

4. An expression involving byte, int, and literal numbers is promoted to which of these?

- ☒ a) int
- ☐ b) long
- ☐ c) byte
- ☐ d) float

5. Which of these literals can be contained in float data type variable?

- ☐ a) `-1.7e+308`
- ☒ b) `-3.4e+038`
- ☐ c) `+1.7e+308`
- ☐ d) `-3.4e+050`

6. Which data type value is returned by all transcendental math functions?

- ☐ a) int
- ☐ b) float
- ☒ c) double
- ☐ d) long

7. What will be the output of the following Java code?

```
class average {  
    public static void main(String args[])  
    {  
        double num[] = {5.5, 10.1, 11, 12.8, 56.9, 2.5};  
        double result;  
        result = 0;  
        for (int i = 0; i < 6; ++i)
```

```

        result = result + num[i];
        System.out.print(result/6);
    }
}

```

- a) 16.34
- b) 16.566666644
- ☒ c) 16.466666666666667
- d) 16.466666666666666

8. What will be the output of the following Java statement?

```

class output {
    public static void main(String args[])
    {
        double a, b,c;
        a = 3.0/0;
        b = 0/4.0;
        c=0/0.0;
        System.out.println(a);
        System.out.println(b);
        System.out.println(c);
    }
}

```

- a) Infinity
- b) 0.0
- c) NaN
- ☒ d) all of the mentioned

9. What will be the output of the following Java code?

```

class increment {
    public static void main(String args[])
    {
        int g = 3;
        System.out.print(++g * 8);
    }
}

```

- a) 25
- b) 24
- ☒ c) 32
- d) 33

10. What will be the output of the following Java code?

```

class area {
    public static void main(String args[])
    {
        double r, pi, a;
        r = 9.8;
        pi = 3.14;
        a = pi * r * r;
        System.out.println(a);
    }
}

```

- ☒ a) 301.5656
- b) 301
- c) 301.56
- d) 301.56560000

Character and Boolean Data Types

11. What is the numerical range of a char data type in Java?

- a) -128 to 127
- b) 0 to 256
- c) 0 to 32767
- ☒ d) 0 to 65535

12. Which of these coding types is used for data type characters in Java?

- a) ASCII
- b) ISO-LATIN-1
- ☒ c) UNICODE
- d) None of the mentioned

13. Which of these values can a boolean variable contain?

- a) True & False
- b) 0 & 1
- c) Any integer value
- ☒ d) true

14. Which of these occupy first 0 to 127 in Unicode character set used for characters in Java?

- ☒ a) ASCII
- b) ISO-LATIN-1
- c) None of the mentioned
- d) ASCII and ISO-LATIN1

15. Which one is a valid declaration of a boolean?

- a) boolean b1 = 1;
- b) boolean b2 = 'false';
- ☒ c) boolean b3 = false;
- d) boolean b4 = 'true'

16. What will be the output of the following Java program?

```
class array_output {
    public static void main(String args[])
    {
        char array_variable [] = new char[10];
        for (int i = 0; i < 10; ++i) {
            array_variable[i] = 'i';
            System.out.print(array_variable[i] + " ");
            i++;
        }
    }
}
```

- ☒ a) i i i i i
- b) 0 1 2 3 4
- c) i j k l m
- d) None of the mentioned

17. What will be the output of the following Java program?

```
class mainclass {  
    public static void main(String args[])  
    {  
        char a = 'A';  
        a++;  
        System.out.print((int)a);  
    }  
}
```

- ☒ a) 66
- b) 67
- c) 65
- d) 64

18. What will be the output of the following Java program?

```
class mainclass {  
    public static void main(String args[])  
    {  
        boolean var1 = true;  
        boolean var2 = false;  
        if (var1)  
            System.out.println(var1);  
        else  
            System.out.println(var2);  
    }  
}
```

- a) 0
- b) 1
- ☒ c) true
- d) false

19. What will be the output of the following Java code?

```
class booloperators {  
    public static void main(String args[])  
    {  
        boolean var1 = true;  
        boolean var2 = false;  
        System.out.println((var1 & var2));  
    }  
}
```

- a) 0
- b) 1
- c) true
- ☒ d) false

20. What will be the output of the following Java code?

```

class asciicodes {
    public static void main(String args[])
    {
        char var1 = 'A';
        char var2 = 'a';
        System.out.println((int)var1 + " " + (int)var2);
    }
}

```

a) 162
☒ b) 65 97
 c) 67 95
 d) 66 98

21. What is the order of variables in Enum?

- a) Ascending order
- b) Descending order
- c) Random order
- ☒ d) Depends on the order() method

22. Can we create an instance of Enum outside of Enum itself?

- a) True
- ☒ b) False

23. What will be the output of the following Java code?

```

enum Season
{
    WINTER, SPRING, SUMMER, FALL
};
System.out.println(Season.WINTER.ordinal());

```

☒ a) 0
 b) 1
 c) 2
 d) 3

24. If we try to add Enum constants to a TreeSet, what sorting order will it use?

- ☒ a) Sorted in the order of declaration of Enums
- b) Sorted in alphabetical order of Enums
- c) Sorted based on order() method
- d) Sorted in descending order of names of Enums

25. What will be the output of the following Java code snippet?

```

class A
{
}

```

```
enum Enums extends A
{
    ABC, BCD, CDE, DEF;
}
```

- a) Runtime Error
- ☒ b) Compilation Error
- c) It runs successfully
- d) EnumNotDefined Exception

26. What will be the output of the following Java code snippet?

```
enum Levels
{
    private TOP,

    public MEDIUM,

    protected BOTTOM;
}
```

- a) Runtime Error
- b) EnumNotDefined Exception
- c) It runs successfully
- ☒ d) Compilation Error